Studying Yb atom dynamics in Optical Dipole traps

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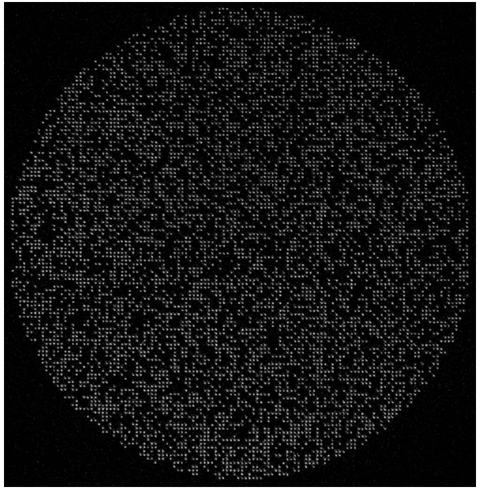
IDDD Project Presentation

Outline

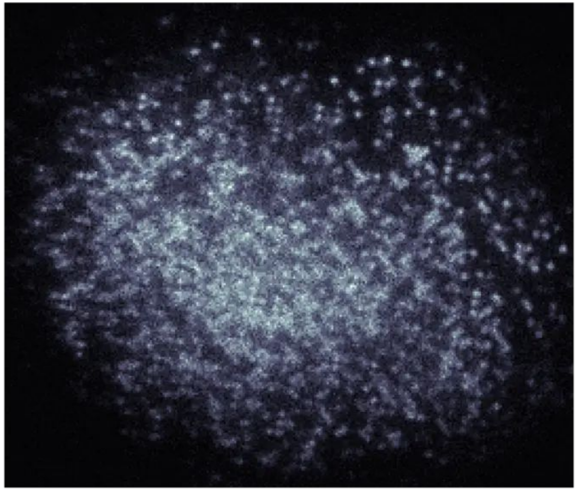
1. Introduction: single-atom imaging
2. Brief recap of mid-term work in 1D quantum dynamics
3. Equivalence of quantum model with classical approach
4. Experimental insights gained, failure modes of classical approach
5. Additional theoretical tools: TDVP
6. Conclusion and next steps

Imaging single atoms in optical traps

- Imaging: scatter photons off an atom to “detect” it
- Used in quantum information/simulation protocols for state readout



Endres Lab @ Caltech:
6000 single atoms in **optical tweezers**



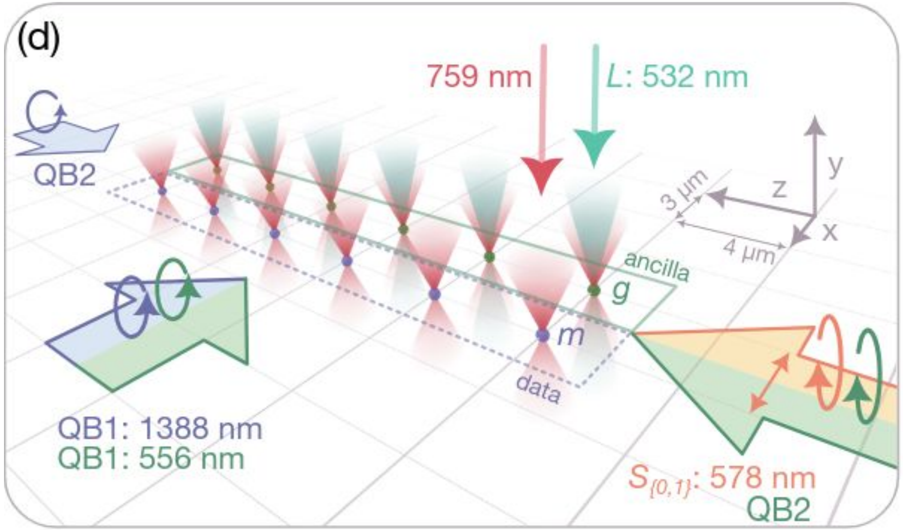
Monika Lab at LMU:
Atoms trapped in an **optical lattice**

Imaging single atoms in optical traps

- Optical traps form potential wells for cold atom to stay trapped.

During imaging, atom scatters photons which causes it to gain Kinetic energy.

Due to finite size of the well, over many such cycles, atoms can effectively get kicked out of the trap, thus limiting readout fidelity.

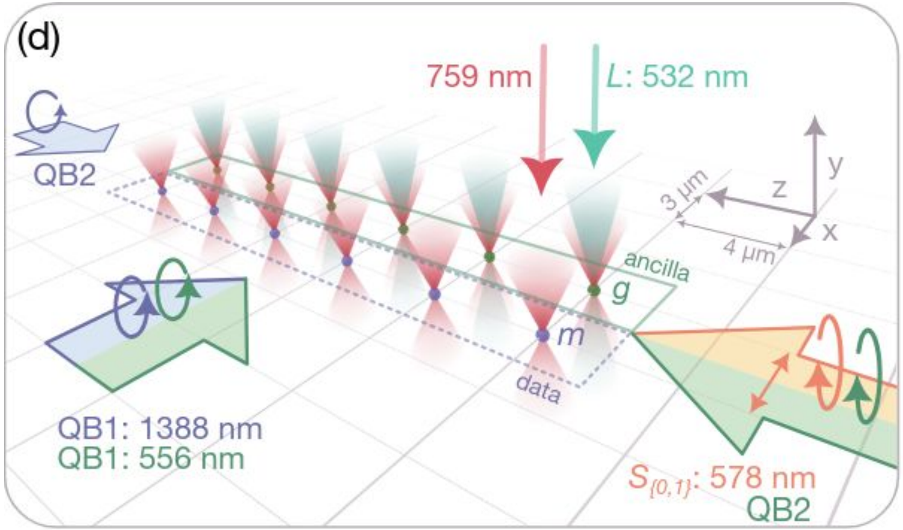


Ref: *arXiv:physics/9902072v1 Grimm et al.*

Imaging single atoms in optical traps

We would like to understand the quantum dynamics of relevant observables during imaging,

and develop an effective model that helps explain experimental observations of atom loss out of a trap.



Ref: arXiv:physics/9902072v1 Grimm et al.

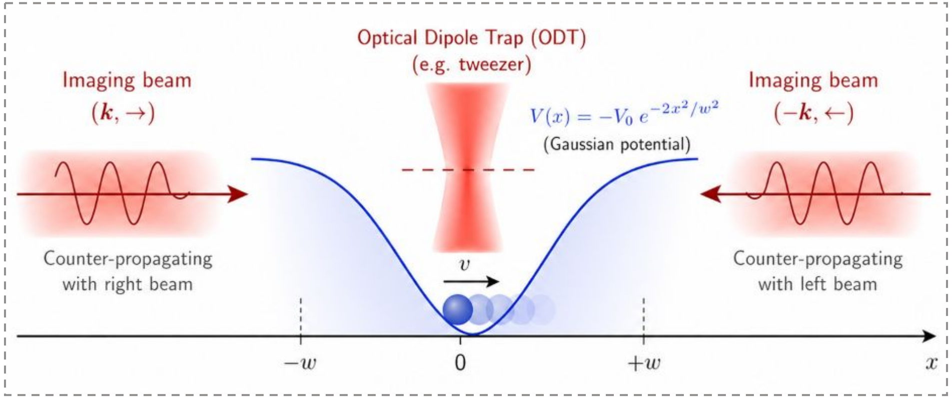
Brief recap: quantum dynamics in a Gaussian well

$$H(x,p) = \frac{p^2}{2m} + V(x) + H_{\text{int}} \qquad L_{\pm} = \sqrt{\frac{\Gamma}{2}}\, |g\rangle\langle e|\, e^{\pm ikx}$$

$$H_{\text{int}} = -\frac{1}{2}\Big[\underbrace{\Omega_1 e^{ikx}}_{+\text{k pulse}} |g\rangle\langle e| + \underbrace{\Omega_2 e^{-ikx}}_{-\text{k pulse}} |g\rangle\langle e| \Big] + h.c.$$

We model the atom being imaged as a TLS interacting with an EM field, with collapse operators modelling SE effects.

Trap freq: ~ 5us
avg. SE rate: ~ every 10ns



Master equation using Quantum Monte Carlo

"Monitor the environment": physically condition evolution on jump/no-jump

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}\left(H_{\text{eff}}\rho - \rho H_{\text{eff}}^\dagger\right) + \sum_{s=\pm} L_s \rho L_s^\dagger$$

$$H_{\text{eff}} = H - \frac{i\hbar}{2}\sum_{s=\pm} L_s^\dagger L_s$$

$$\langle A \rangle_t \approx \frac{1}{N}\sum_{i=1}^{N} \frac{\langle \psi_i(t)|A|\psi_i(t)\rangle}{\langle \psi_i(t)|\psi_i(t)\rangle}$$

1. Non-unitary time evolution of $|\psi\rangle$ under H_{eff} (causes norm decay)
2. SE with probability $\propto \langle \psi | L_s^\dagger L_s | \psi \rangle$

If jump: $|\psi\rangle \rightarrow \dfrac{L_s|\psi\rangle}{\|L_s|\psi\rangle\|}, \quad L_s \sim \sqrt{\Gamma}\sigma_{ge}$

No jump: $|\psi\rangle \rightarrow \dfrac{e^{-iH_{\text{eff}}dt/\hbar}|\psi\rangle}{\|e^{-iH_{\text{eff}}dt/\hbar}|\psi\rangle\|}$

Ref: arXiv:1405.6694 *Daley et al.*

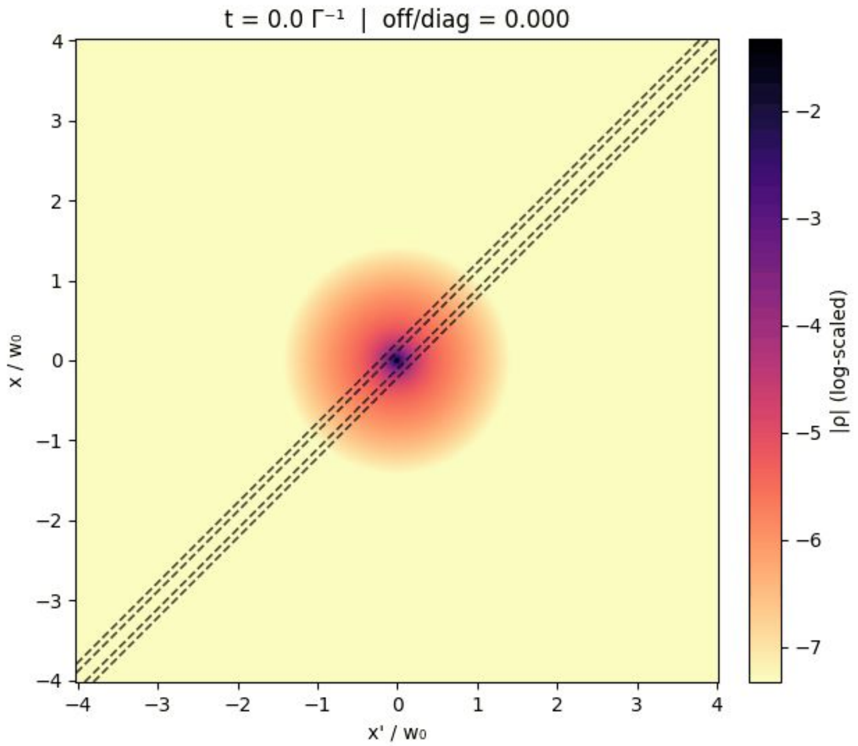
1D QM density matrix

We find an interesting result:

The positional density matrix remains ~diagonal and well-localised within a length scale set by wavenumber of the emitted photon:

~ 1/k ~ 60nm

(Pichler et al, PRA 2010)



Since our dynamics takes place in a trap of width 500nm (>> 1/k), this motivated us to compare results against a classical stochastic description of dynamics.

1D semi-classical model

Core idea: replace quantum wavefunctions with classical point particles undergoing hamiltonian evolution interrupted by stochastic kicks.

$$\text{Strong driving: } \Gamma_{\mathrm{sc}} = \Gamma/2$$
$$t_{\mathrm{sc}} \sim \mathrm{Exp}(2/\Gamma)$$

$$p \;\to\; p + \Delta p, \quad \Delta p \in \{\pm \hbar k_{\mathrm{abs}} \pm \hbar k_{\mathrm{em}}\}$$

Stochastic update at exponentially sampled times

$$\dot{x} = \frac{p}{m}, \qquad \dot{p} = -\frac{dV(x)}{dx}$$

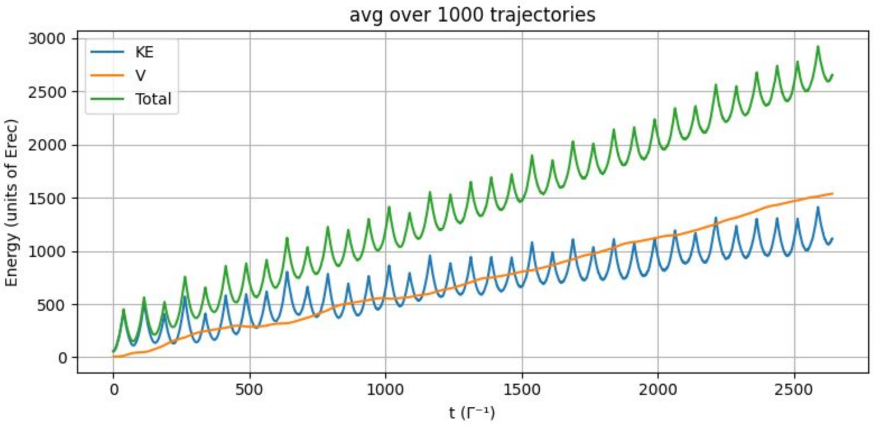
$$V(x) = -V_0 e^{-x^2/(2\sigma^2)}$$

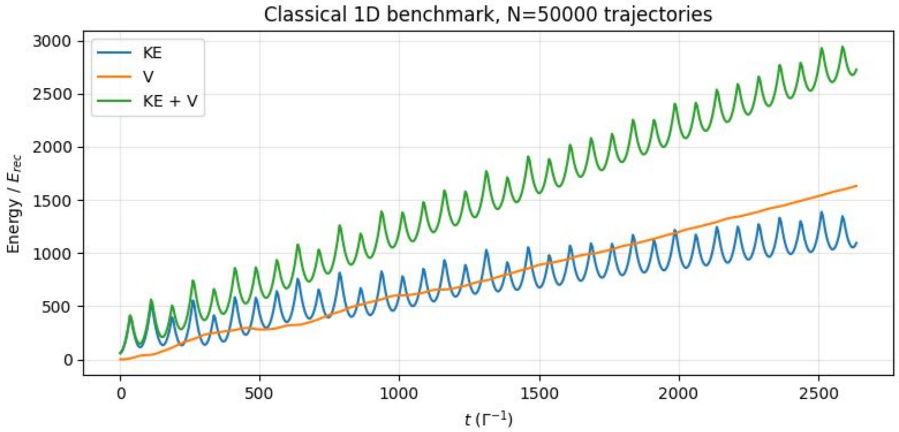
$$\dot{p} = -\left(\frac{V_0 x}{\sigma^2}\right) e^{-x^2/(2\sigma^2)}$$

Hamiltonian update

1D semi-classical vs QM comparison

Closely agree over timescales of 20us; **doppler effects** cause slight divergence at longer times

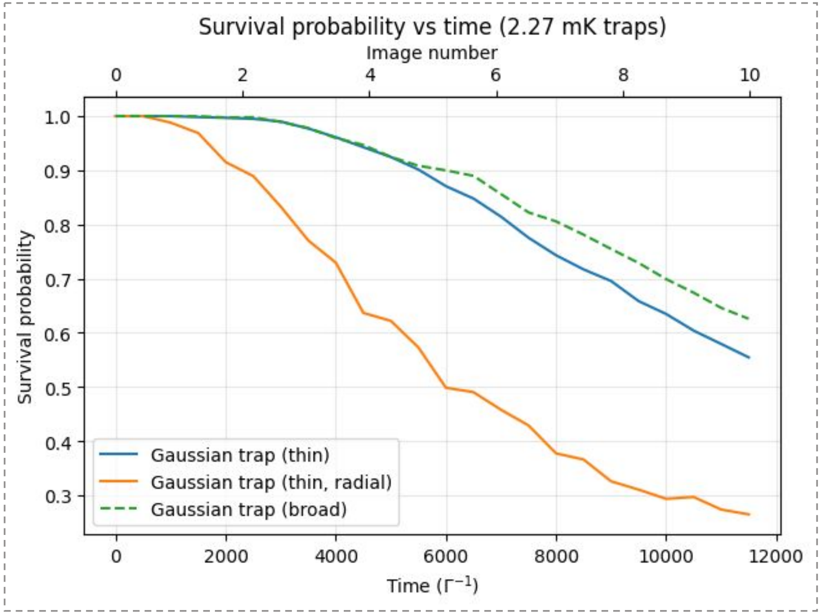


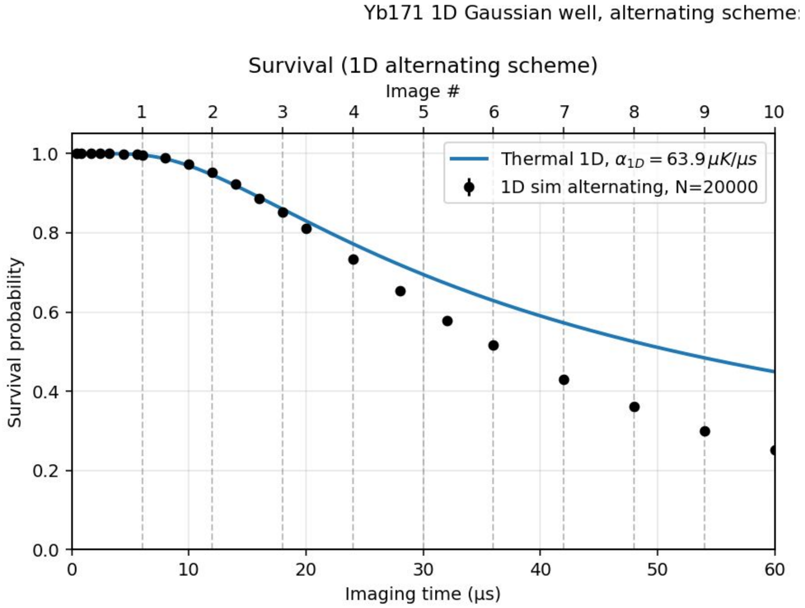


V > KE: atom begins to explores anharmonic part of the trap

1D semi-classical vs QM comparison

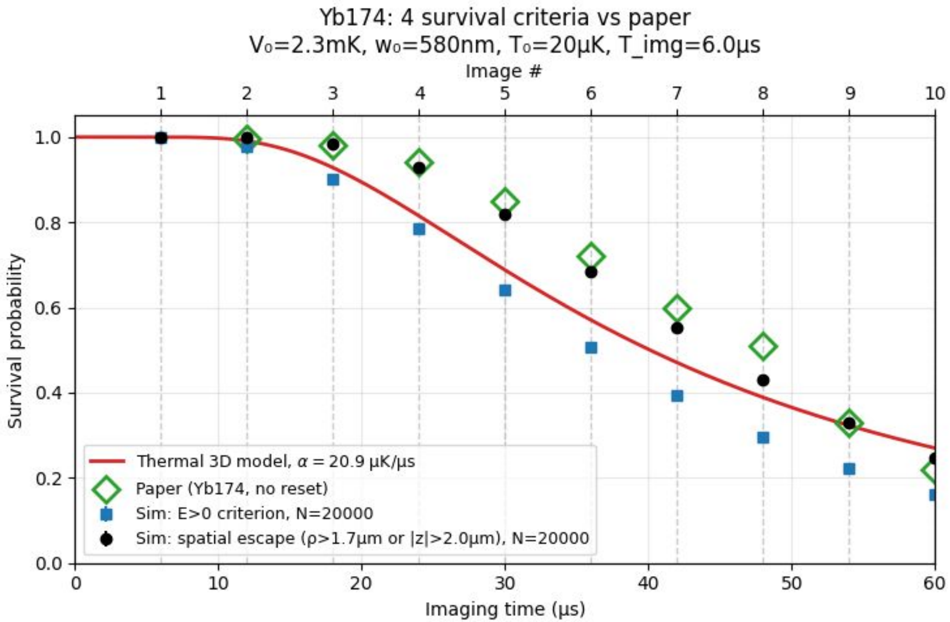
Loss model: P(E > 0) gives excellent agreement between QM and classical predictions.





Atom in a 3D trap: experimental prediction

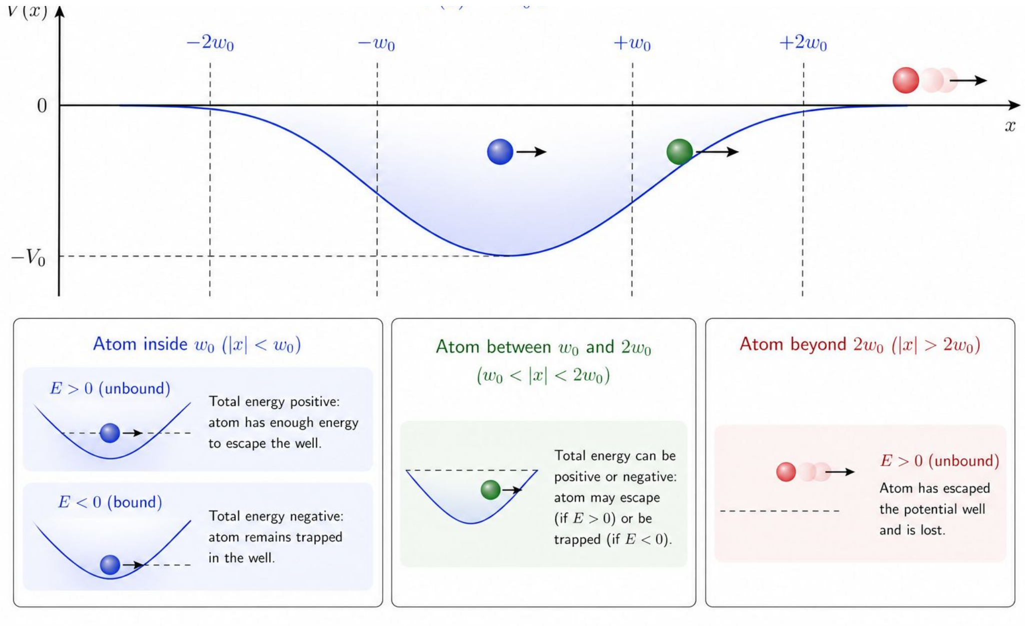
Using this model in 3D, we were able to explain long-term atom loss better than the model proposed by the authors of experimental data.



This also helped us redefine our interpretation of a “lost” atom to beyond just <E> > 0,

because experimentally there are atoms with <E> > 0 that survive because they are localised!

Atom loss explanation



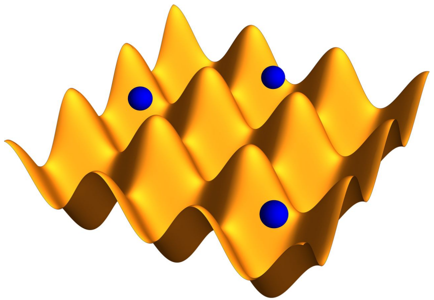
Energetic atoms (blue) liable to be lost, but not in the timescales of imaging

=> Can be recaptured using cooling schemes, thus improving atom survival.

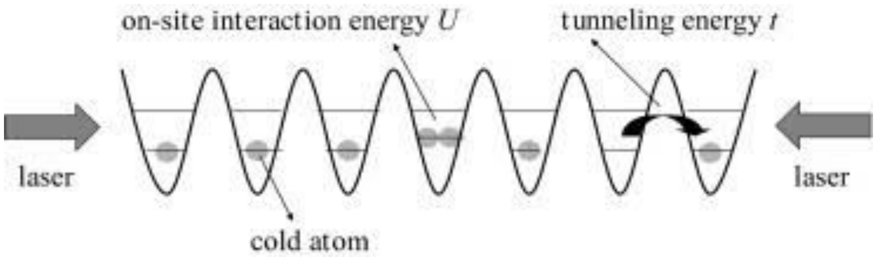
For example: for an imaging duration of 10us, of the **2% atoms with <E> > 0**, **1% are localised within the well**, and hence are not effectively “lost”.

Failure modes of classical stochastic approach

- stochastic model works when: short timescales, fast driving/dissipation
- Same conditions may not hold for other traps such as **optical lattices**, where tunneling effects form a major atom loss mechanism.
- This calls for new approach for 2/3D quantum dynamics, one that can also capture tunneling effects between potentials.



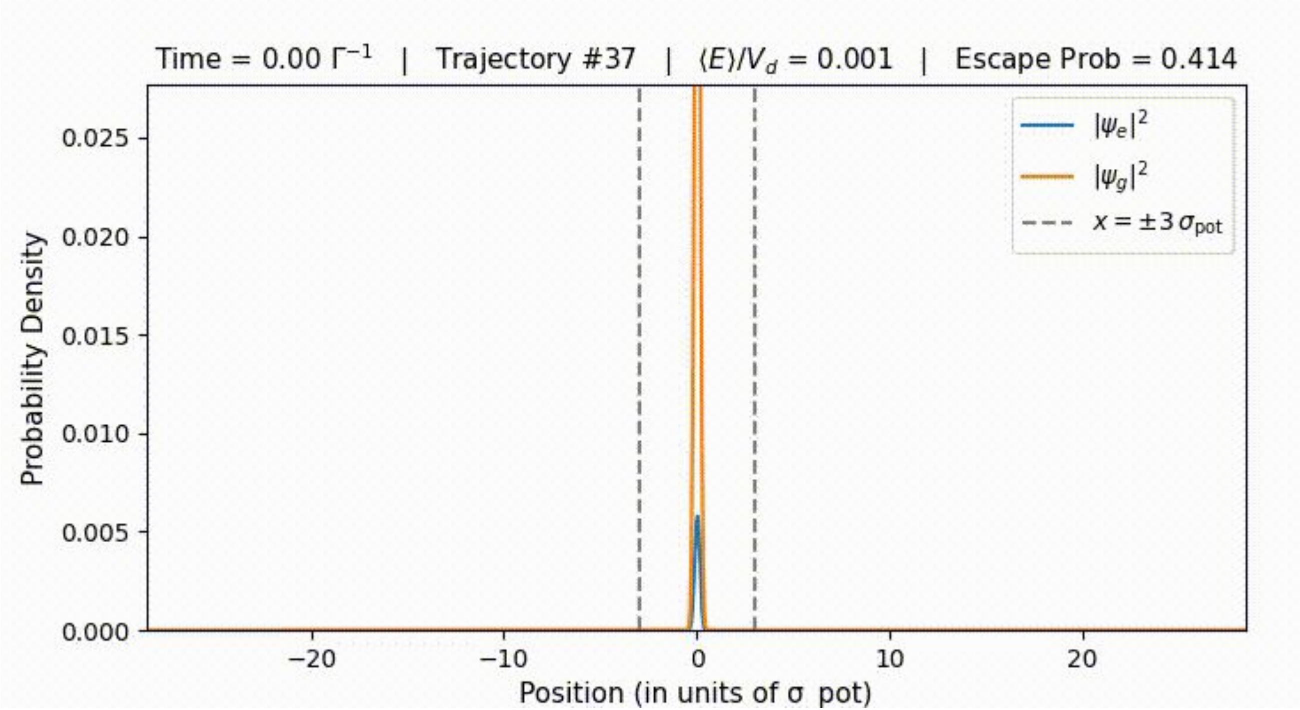
Ref: wikipedia



Ref: sciencedirect

Quantum dynamics using adaptive basis: TDVP

1D sim: wavepacket closely follows center-of-mass motion in the well



Quantum dynamics using adaptive basis: TDVP

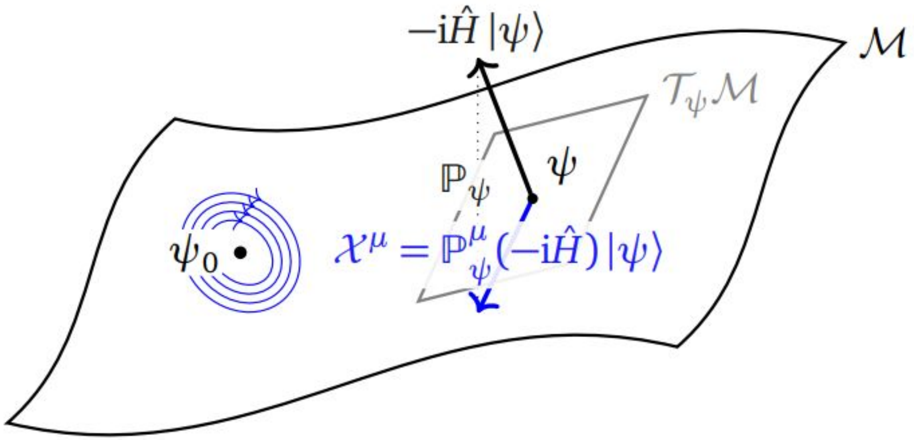
- Dynamics in a well: COM motion + local Hilbert space to capture quantum effects

- Achievable by evolution using **time-dependent variational principle (TDVP)**

Key idea:

We restrict |psi> to lie on a "manifold"

$$|\psi_{\mathrm{true}}\rangle \longrightarrow |\psi(z)\rangle$$



Ref: SciPostPhys 9.4.048

Quantum dynamics using adaptive basis: TDVP

In this framework, the Schrodinger equation transforms into an algebraic form:

$$i\partial_t \left|\psi_{\text{true}}\right\rangle = H \left|\psi_{\text{true}}\right\rangle$$

$$\downarrow$$

$$g_{\mu\nu}(\boldsymbol{z})\, \dot{z}^{\nu} = -i\langle\partial_\mu\psi|H|\psi\rangle + \text{c.c.}$$

For our problem, we have used a **sum-of-Gaussian states** ansatz for the wavefunction.

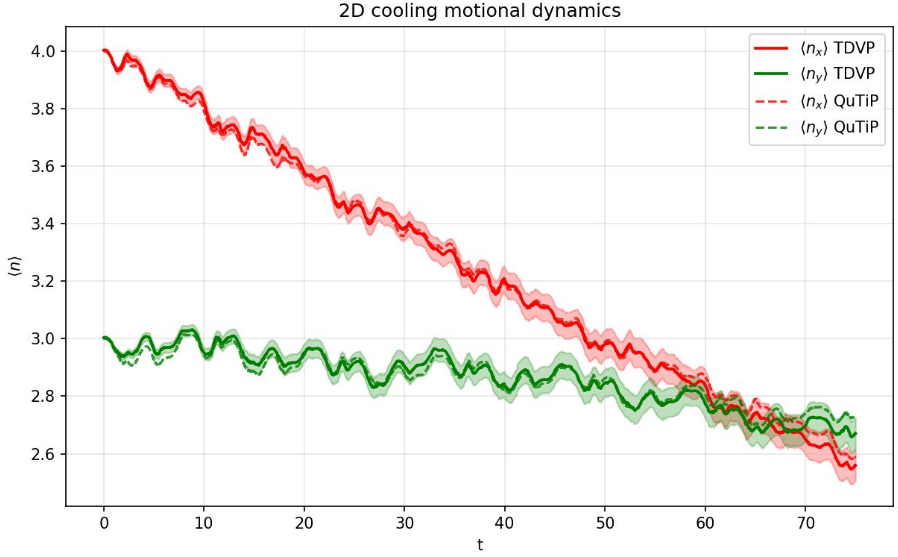
$$\boxed{\psi = \sum_k c_k\, D(\alpha_k)\, S(\beta_k)\, |0\rangle}$$

TDVP for multimode problems

Following similar extension to quantum MC as before, we have demonstrated its applicability to studying non-LD regime sideband cooling of atoms in 1D and 2D QHO.

QuTiP: N_fock = 35 per mode.
TDVP: 6 Gaussians.

For 2(+)D cooling problems, we observe significant runtime advantage using TDVP.



Conclusions

- Successfully described dynamics for atoms in optical tweezer potential, using a semiclassical model
- Motivated applicability of semiclassical model using an exact 1D QM result
- Developed and tested a new numerical technique (TDVP) for 3D dynamics

Next steps:

- Benchmark our 3D result for tweezers using TDVP **(in progress)**
- Exact studies of losses in optical lattice, characterize over the barrier (hopping) vs under the barrier (tunneling) losses **(near future)**
- Continue building experiment at MPQ to test theoretical predictions for lattice.

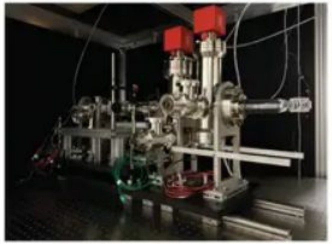
Experimental work

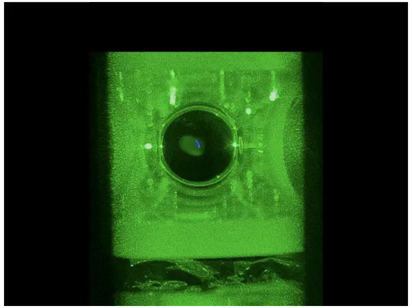
Timeline

Aug ‘25

Nov ‘25 (my stay at MPQ) May 2026
(We are HERE)

September/October ‘26





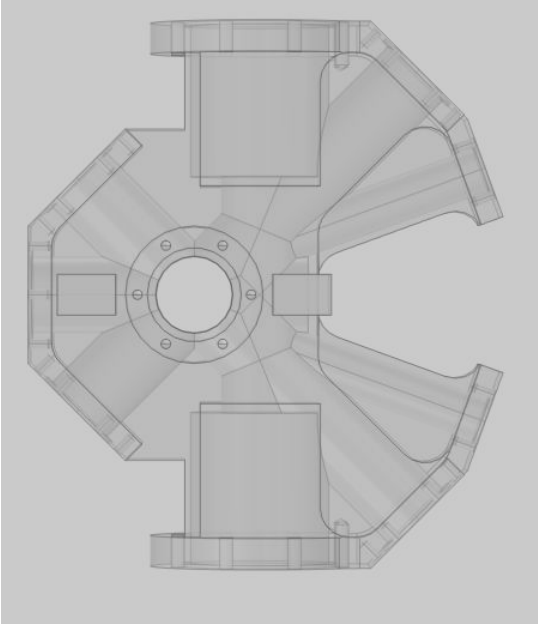
New Lab

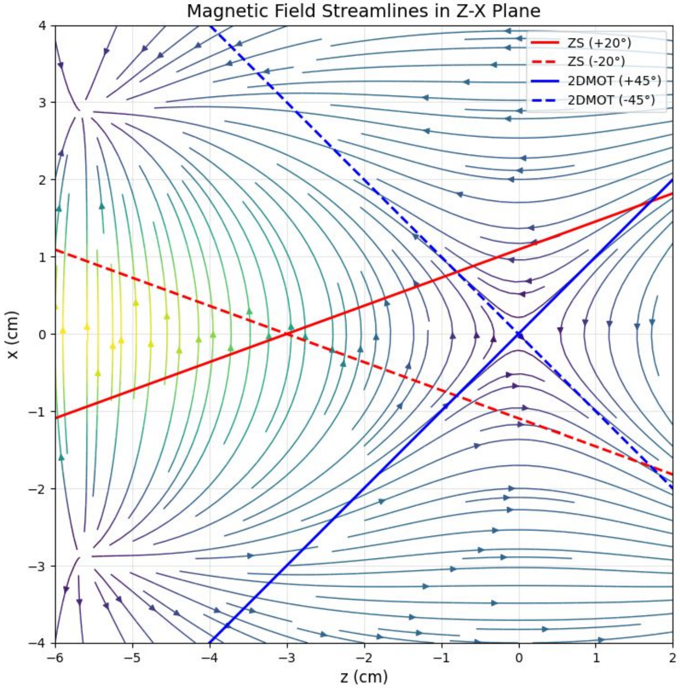
New Yb Atomic source +
3D Magneto-Optic trap

Single atoms trapped in
optical lattices (soon)

My contributions to the expt.

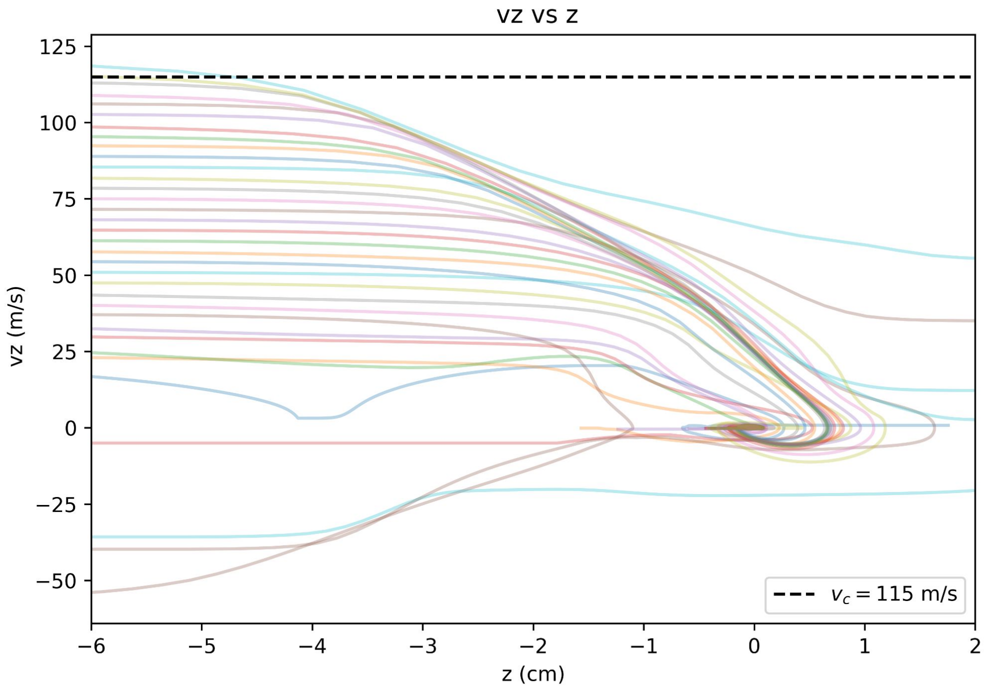
Main project: **designing a Zeeman Slower and 2D-Magneto Optic Trap**





My contributions to the expt.

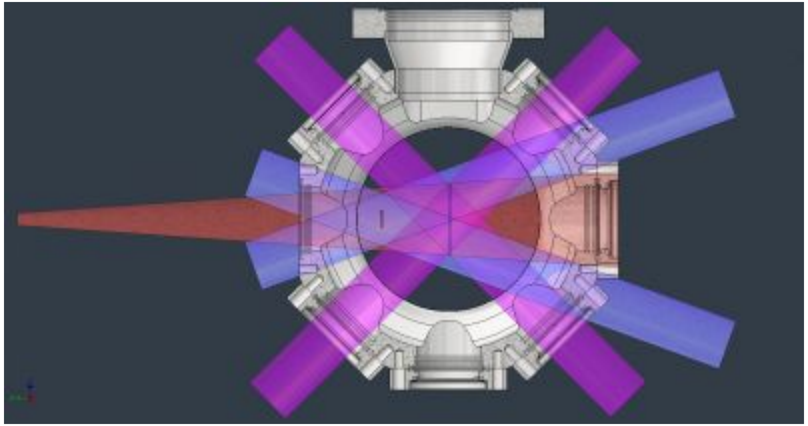
Semiclassical simulations estimating capture velocity of atomic source

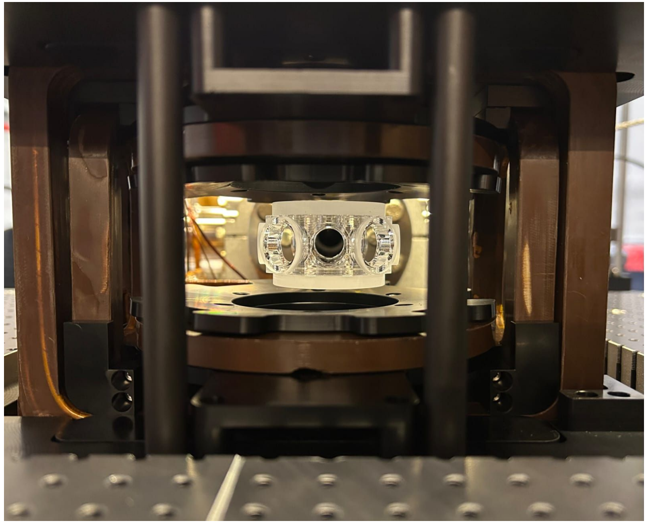


Along with a postdoc and a PhD student, we iteratively designed the aforementioned chamber and the rest of the source using information from such simulations

To maximize atomic flux that results.

Currently awaiting components!





Thank you!